

Taller # 1 de fisicoquímica.

1. What is a pneumatic trough and why was its invention important?
2. Who invented the barometer? What unit bears his name?
3. What determines the pressure of a gas?
4. What is the general statement of Boyle's law?
5. For any gas at constant temperature what is the equation that describes Boyle's law?
6. What is the SI unit for pressure? What is the standard scientific unit for pressure?
How they interconverted?
7. How is the pressure unit torr related to the pressure unit atmospheres?
8. What is the kinetic-molecular theory of gases?
9. What are the freezing and boiling points of water in degrees Celsius? Degrees Fahrenheit?
10. What is the conversion factor between degrees Fahrenheit to degrees Celsius?
11. What is the conversion factor between degrees Celsius and degrees Fahrenheit?
12. Why are the Celsius and Fahrenheit temperature scales said to be empirical?
13. What is the temperature of absolute zero in degrees Celsius?
14. What is thought to happen at absolute zero?
15. Why is the Kelvin temperature scale an absolute scale?
16. What is the conversion factor between degrees Celsius and degrees Kelvin?
17. What is the absolute temperature scale used in some engineering fields?
18. How does it differ from the Kelvin scale?
19. What is the general statement of Charles' law?
20. What is the equation that describes Charles' law?
21. What is the general statement of Gay-Lussac's law?
22. What is the equation that describes Gay-Lussac's law?
23. What is the equation of the combined gas law?
24. What is the general statement of Avagadro's law?
25. What is the equation that describes Avagadro's law?
26. What is the equation of the ideal gas law?
27. What is an ideal gas?
28. What is the value of the gas constant in $\text{L}\cdot\text{atm}/\text{K}\cdot\text{mol}$?
29. Under what conditions do gases not behave ideally?
30. What are the three assumptions of the kinetic-molecular theory of gases that do not hold at low temperatures and high pressures?
31. What is the van der Waals equation?
32. What do the van der Waals constants correct for?
33. Which van der Waals constant is most important for polar molecules? Which constant is more important for larger molecules?
34. What is the statement of Dalton's law of partial pressures?
35. What is a mole fraction?
36. What is STP?
37. What is the volume of one mole of a gas at STP?
38. What is the law of combining volumes?

- 38 Convert the following pressures to atmospheres, (a) 1.4×10^5 Pa, (b) 2.53 bar, (c) 1520 Torr, (d) 19.1 psi, (e) 836 mmHg, (f) 507 millibar.
- 39 If the volume of gas in a piston is 5.63 L at a pressure of 1.5 atm and the gas is compressed to a volume is 4.8 L, what will be the new pressure inside the piston if the temperature remains constant?
- 40 A gas occupies 12.3 L at a pressure of 40.0 mmHg. What is the volume when the pressure is increased to 60.0 mmHg at constant temperature?
- 41 A 1.5 L container is filled with nitrogen at a pressure of 10 atm. What size container would be required to hold this gas at a pressure of 2.0 atm at the same temperature?
- 42 A gas occupies 1.56 L at 1.00 atm. What will be the volume if the temperature remains the same and the pressure is increased to 3.00 atm?
- 43 If the pressure on a gas is decreased by one half, how will the volume change if the temperature does not?
- 44 A weather balloon contains 15 L of air at sea level. What will be the volume of the balloon when it ascends to an altitude where the air pressure is 0.85 atm at the same temperature?
- 45 A gas occupies 77.0 L at 18.0 Torr at standard temperature. What will its volume be at STP?
- 46 Convert the following temperatures to Fahrenheit: (a) 15.5°C, (b) 25°C, (c) 52.3°C, (d) 75°C, (e) 83°C.
- 47 Convert the following to Celsius: (a) 167°F, (b) 131°F, (c) 86°F, (d) 298 K, (e) 324 K.
- 48 Convert the following to Kelvin: (a) 25.5°C, (b) 50.3°C, (c) 98.0°C, (d) 256°R, (e) 536°R.
- 49 What is the decrease in temperature when 2.00 L of a gas at 25.0°C is compressed to 1.00 L if the pressure remains the same?
- 50 What is the volume of a gas at 100°C and standard pressure if it occupies 1.00 L at STP?
- 51 A 5.00 L sample of gas is collected at 100 K and then allowed to expand to 20.0 L. What is the new temperature of the gas if the pressure does not change?
- 52 What is the volume of a gas sample at 150°C if it has a volume of 6.00 L at 27.0°C at the same pressure?
- 53 A balloon has a volume of 2500 mL during the day at a temperature of 28.8°C. If the temperature at night falls to 10.0°C, what will the volume of the balloon be, in liters, if the pressure remains constant?
- 54 The volume of a gas in a sealed cylinder is 30 cm³ at a temperature of 30°C. As the gas is heated to 60°C, the piston moves to maintain the same pressure. What is the new volume of the gas in liters?
- 55 If 50.0 L of oxygen at 20.0°C is compressed to 5.00 L, what must the new temperature be to maintain constant pressure?
- 56 What is the final pressure of a gas sample in a rigid container at 1.00 atm when it is heated from 25.0°C to 40.0°C.

- .57 What is the final pressure, in standard atmospheres, of a gas sample in the steel tank at 3000 Torr when the gas is cooled from 25.0°C to 0.00°C.
- .58 A gas has a pressure of 640 Torr at 40.0°C. What is the temperature at standard pressure?
- .59 A gas is collected in a rigid container at 22.0°C and 745 Torr. If the temperature decreases to 0.0°C, what is the pressure in the container in atmospheres?
- .60 A gas sample is collected in a rigid cylinder under 1.25 atm pressure. At 36.5°C, that same gas sample has a pressure of 2.50 atm. What was the initial temperature of the gas in the cylinder?
- .61 A gas sample has a volume of 17 L, a pressure of 2.3 atm, and a temperature of 299 K. If the temperature of the gas is raised to 350 K and the pressure is lowered to 1.5 atm, what is the new volume of the gas sample?
- .62 A gas sample has a volume of 17 L, a temperature of 67°C, and a pressure of 88.89 atm, what will be the pressure of the gas sample if the temperature is raised to 94°C and the volume is decreased to 12 L?
- .63 A 2.00 L gas sample is collected at 25.0°C and 745.0 Torr. What is the volume of the sample at STP?
- .64 The volume of a gas sample at STP is 500 mL. What volume would the same gas occupy when at a pressure of 3.00 atm and temperature of 100°C?
- 65 If the absolute temperature of a gas sample is doubled and the pressure tripled, how does the volume change?
- 66 If 4 mol of a gas is contained in a 12 L container at a pressure of 5.6 atm, what is the temperature of the gas?
- 67 What is the volume of 20.3 g of carbon dioxide gas at STP?
- 68 How many moles are there in a gas sample at STP and a volume of 56.2 L?
- 69 What is the molecular weight of a 64.1 g sample of gas that occupies 22.4 L at STP?
- 70 What is the molecular weight of a gas that occupies 19.2 L at STP with a mass of 12.0 g?
- 71 What is the pressure of 2.4 mol of a gas sample in a 45 L container at a temperature of 97°C?
- 72 What is the density of ammonia gas at 1.00 atm and 298 K?
- 73 What is the molar mass of a gas that has a density of 1.853 g/L at 745.5 Torr and 23.8°C?
- 74 Which of the following gases has the smallest van der Waals "a" constant: NH₃, N₂, H₂O, CCl₄?
- 75 Which of the gases in **Problem 6.74** has the largest van der Waals "b" constant?
- 76 Compare the calculated pressure of 1.00 mol of carbon dioxide gas in a 536 mL container at 373 K using: (a) the ideal gas equation and (b) the van der Waals equation ($a = 3.64 \text{ L}^2 \cdot \text{atm} \cdot \text{mol}^{-2}$; $b = 0.0427 \text{ L/mol}$).
- 77 What is the deviation from ideal behavior for carbon dioxide under the conditions of **Problem 6.76**?
- 78 What is the total pressure of a mixture of 2.00 g of hydrogen, 8.00 g of nitrogen, and 12.0 g of argon in a 10.0 L container at 273 K?
- 79 A mixture of 1.01 g of hydrogen and 17.7 g of chlorine in a container at 300 K has a total gas pressure of 0.975 atm. What is the partial pressure of hydrogen in the mixture?
- 80 A 1.00 L flask contains 1.10 g of carbon dioxide and an unknown amount of oxygen at 373 K and 608 Torr. What is the mass of oxygen in the flask?
- 81 A tank contains 480 g of oxygen and 80 g of helium at a total pressure of 7.0 atm. (a) What is the mole fraction of oxygen in the tank? (b) What is the partial pressure of oxygen?
- 82 What is the amount of oxygen produced from the decomposition of 4.00 g hydrogen peroxide ($\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$)?
- 83 If 2.0 mol of methane (CH₄) are completely combusted at STP, how many liters of CO₂ are produced (see [Section 4.4](#))?
- 84 Ozone decomposes in water to yield hydrogen and oxygen gas. What volume of ozone is required to produce 5 L of oxygen gas at STP?
- 85 Ammonia gas reacts with oxygen to yield NO₂ and H₂O. (a) What is the balanced equation for this reaction? (b) What volume of NO₂ is produced from 100 g of NH₃(g) at STP?